Hardware-Based Anti-Counterfeiting Techniques for Safeguarding Supply Chain Integrity

Md Tanvir Arafin
ECE Department
University of Maryland
College Park, Maryland
Email: marafin@umd.edu

Andrew Stanley
CISO
Philips
Andover, Massachusetts
Email: andrew.stanley@philips.com

Praveen Sharma
Cyber Innovations Team
Philips
Andover, Massachusetts
Email: praveen.sharma@philips.com

***Abstract*—Counterfeit integrated circuits (ICs) and systems have emerged as a menace to the supply chain of electronic goods and products. Simple physical inspection for counterfeit detection, basic intellectual property (IP) laws, and simple protection measures are becoming ineffective against advanced reverse engineering and counterfeiting practices. As a result, hardware security-based techniques have emerged as promising solutions for combating counterfeiting, reverse engineering, and IP theft. However, these solutions have their own merits and shortcomings, and therefore, these options must be carefully studied.**

In this work, we present a comparative overview of available hardware security solutions to fight against IC counterfeiting. We provide a detailed comparison of the techniques in terms of integration effort, deployability, and security matrices that would assist a system designer to adopt any one of these security measures for safeguarding the product supply chain against counterfeiting and IP theft.

I. Introduction

Hardware counterfeiting of enterprise products is increasingly becoming one of the fastest growing criminal ventures [1]. The electronic design and manufacturing industries are vulnerable to these counterfeiting attacks. Product counterfeiting ranges from simple relabeling attacks to extensive reverse engineering and fabrication ventures. Not only do the commodity products fall victim to such criminal activities, but also a product's subcomponents, major system components, and critical integrated circuits (ICs) in the system are reverse engineered and counterfeited. As a result, IC counterfeiting is becoming a serious concern for supply-chain integrity, a menace for the security of a business entity, and a threat to public health, safety, and national security. In response to these threats, active research is being pursued designing and advancing current electronic manufacturing and distribution processes for preventing IC counterfeiting and reverse engineering efforts.

Counterfeit ICs have the potential to become pervasive in critical electronic devices. They have found their ways in medical devices such as Automated External Defibrillators (AEDs) and intravenous drip machines, critical infrastructures such as braking systems in high-speed trains, and power supplies in airport landing lights, and even in radiation detectors and nuclear submarines [2]. Advances in integrated circuit design and fabrication technology have reduced the cost of manufacturing and reverse engineering tremendously. As a result, of IC counterfeiting has emerged as a lucrative business model, resulting in a billion dollar electronic circuit-counterfeiting trade around the globe. Some areas impacted by illegally manufactured and distributed ICs are as follows:

a. *Loss in revenue, intellectual property, and brand recognition*: Counterfeited ICs usually replace a known and trusted manufacturer, which can result in a tremendous loss in revenues for the legitimate manufacturer. Furthermore, counterfeiting is an attack on the brand recognition and reputation of the business, and therefore, it can affect brand in the market competition in the long run. In 2011, Semiconductor Industry Association (SIA) estimated at $7.5 billion cost incurred per year to the electronic industries due to IC counterfeiting [2].

b. *Loss of reliability in a critical system's functional capability*: Counterfeit components can find their way into critical systems such as military equipment, financial infrastructure, electronic grids, transportation systems, and healthcare products. In turn, counterfeited electronics have the potential to create major reliability and security issues for critical systems and infrastructure.

c. *Malware injection*: One of the common by-products of counterfeit electronic goods is the proliferation of malware in a system containing such electronic products. For example, malicious executables are commonly reported in forged memory components [2].

d. *Hardware trojan insertion*: Since illegal manufacturers can reverse engineer or alter hardware components, they can easily insert hardware trojans. Such malicious alterations create targeted attacks on a system in the future, or can leak critical information or cryptographic keys to an adversary [3].

Counterfeit ICs can enter into the supply-chain at different points as shown in Fig. 1. Therefore, rigorous supply chain management is required for combating against fake electronic circuits. One of the simplest solutions to this problem is incorporating Certification of Authenticity (CoA) measures with the products and their electronic components. However, maintaining such effort can be difficult due to human factors. Since we are dealing with hardware counterfeiting, software-based authentication is inadequate. If the counterfeit product is capable of executing similar systems as the original one, then it is difficult to differentiate them using software-based solutions such as functional testing. Therefore, in this work, we examine hardware security solutions for anti-counterfeiting techniques.

978-1-4673-6853-7/17/$31.00 ©2017 IEEE

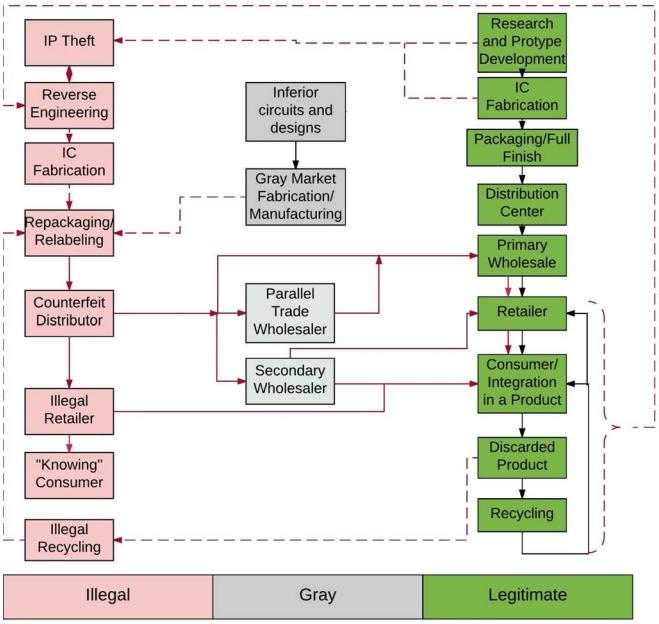

Fig. 1. Paths for counterfeit ICs to enter into the IC supply chain. The green boxes represent legitimate steps in the life-cycle of an IC, the gray boxes represent the gray market players that affect the supply chain, and the red boxes represent the life-cycle of a counterfeit IC. The solid red lines represent the paths that an illegal design uses to infiltrate the supply chain. The dashed lines show ways that a legitimate design and supply process help sustain an illegal counterfeiting enterprise. Similar designs can be found in [4].

II. Counterfeiting Techniques

The prevailing belief about counterfeiting is that it is usually done by adversaries with limited manufacturing capacity and that their business model is notional. This general notion can explain most of the common practices in counterfeiting, such as repackaging and reselling an inferior product as its superior counterpart, and recycling an older chip and relabeling it as a new one. Such techniques obviously do not require advanced manufacturing capabilities; however, this business model relies on the supply of cheaper alternatives of a given product in the market for sustaining a profit margin. This exposes a gray market, which constitutes the supplier of these substandard products and the counterfeiters. The manufacturers of these low-grade electronic products have a capacity for reverse engineering and manufacturing a given design. From a security perspective, this creates a more serious concern because such manufacturers can insert hardware or software trojans in the design, which can find its way inside a secure system.

We can categorize the IC counterfeiting techniques into the three common categories discussed below.

A. Relabeling and Repackaging

This is the simplest attack on the supply chain of a given electronic product. Inferior, defective, or overproduced ICs are relabeled and repackaged for monetary gain. For example, in recent years there has been an increased number of relabeling attacks on memory components such as flash drives and SD cards used in the commodity electronics. There are two ways these attacks are being performed. Firstly, a lower capacity memory component is relabeled as a higher capacity one. In such cases, there are reports of malicious software and hardware controllers, which misrepresent the capacity of the component to the system and erase data automatically to give the illusion of larger capacity [5]. The second type of counterfeiting involves using a low-grade component repackaged as some higher grade one. For example, an SD card without ECC is repackaged as one with ECC [5].

B. Reverse Engineering and Illegal Manufacturing

To maintain a supply of low-grade parts and ICs, manufacturing is essential. Such manufacturing involves stealing intellectual properties by reverse engineering a given electronic product. With the advancement of probing and testing technologies, reverse engineering has become more accessible than before. Furthermore, the manufacturing cost for electronic components is getting cheaper. As a result, reverse engineering simpler IC designs and components and manufacturing them for gaining market share is becoming a rewarding business model for counterfeiters and gray market entrepreneurs.

C. Illegal Recycling

Another simple method of counterfeiting is recycling older chips. This weakness comes from the consumer side of the supply chain. Usually, at the end of a product lifecycle, consumers discard the product with some of the components functioning. Unregulated disposal of aged products has led to the business of recycling older ICs and selling them as newer ones. Since the IC primarily originates from a trusted source, recycling can retain system security; however, this practice can create serious reliability concerns for a given system.

III. Hardware Security Measures for Safeguarding Supply Chain

As discussed in the previous sections, human factors can weaken the integrity of the supply chain of a given chip design. Law enforcement and subsequent legal actions can address IC counterfeiting problems; however, legal actions are inherently an after-the-fact measure. Therefore, necessary prevention techniques must be adopted to stop the proliferation of the counterfeited IC before or during attacks to the supply chain.

Physical inspection can be a simple solution for counterfeit detection. However, with increased capabilities and cheaper printing and packaging techniques, it will become increasingly difficult to detect a counterfeit IC. Therefore, in this work, we compare hardware security-based techniques that provide better solutions against counterfeiting. Hardware security measures can provide two key benefits:

1. Dynamic detection of counterfeited ICs and prevention of further damage in the system
2. Legal help to the authorities for finding the details of the supply chain of illegal goods and for pursuing IP infringement cases

However, introducing these security measures incurs cost, and therefore, a designer must weigh the options carefully to better suit the target product and the nature of counterfeiting. This work presents simple comparisons between different anti-counterfeiting techniques to support a hardware designer. Overall, in this work, we have measured different existing hardware-based anti-counterfeiting solutions in terms of three parameters:

A. Integration Effort

E1. Design overhead (power/area/cost)

E2. Design effort and cost of intellectual property

E3. Effect on yield and lead-time

E4. Maturity of the technique

B. Deployability

D1. Deployment cost per user

D2. Maintenance cost per user

D3. Cost of recycling

D4. Ease of use

D5. Ease of testing

D6. Backwards compatibility

C. Security

S1. Resiliency to testing attacks

S2. Resiliency to reverse engineering

S3. Probability of success of exploitation

S4. Ease of counterfeit detection

S5. Ease of recovery from loss

S6. Effectiveness in IP protection

One of the simplest solutions is tagging a serial number on electronic components and keeping a record of the valid tags. We compare all of the hardware security solutions against this simple measure of counterfeit protection. It should be noted that the application of any one of these technologies would depend on the nature and spread of a counterfeiting problem and the product. Therefore, we do not rank the technologies by their average merit; instead, we present their effectiveness in different scenarios, which would give a designer a practical overview of which technique to incorporate into his design. A graphical representation of this comparison is given in Fig. 2.

Fig. 2. Comparison of different anti-counterfeiting techniques

T1. *Tagging and Certification of Authenticity*: A common form of anti-counterfeiting technique is tagging the packaging of an authentic chip with a unique serial number and keeping a database of all authentic chips. A Certificate of Authenticity can be provided for all chips carrying an authentic serial or identification tag. For implementation, this approach does not require additional design overhead in terms of power or the area budget of the chip. This technique is easily deployable too since the only cost of deployment is in maintaining the database of the authentic product. However, this approach does not provide any security against reverse engineering and IP theft. Furthermore, a counterfeiter can easily copy the tag and relabel a fake IC. Therefore, this method does not fare well in terms of security.

T2. *Watermarking*: To identify the sources of IP theft, hardware-based watermarking techniques can be used. Watermarking adds signatures in the circuit design, which is unique for a given instance of a bulk production. Watermarks are added on a given IP to create multiple versions of the IP for distribution for manufacturing at different times or at different facilities. Hardware watermarks do not change the functionality of the IC, rather they add information about the distribution of a given IP [6]. In the event of an IP theft, the available counterfeited ICs can be examined to reveal the source of the manufacturer from which the IP was stolen. Watermarking is relatively easy to implement and deploy, however, it provides limited assistance in counterfeit detection and safeguarding the supply chain.

T3. *Fingerprinting*: While watermarking creates a unique signature for each batch of fabricated devices, hardware fingerprints are unique for each device. These fingerprints can be added by incorporating unique keys in each IC, or by exploiting fabrication variation. Hardware fingerprinting techniques are based on the idea that each device is capable of creating unique fingerprints due to the incorporation of fingerprint generators, combinational logic locks [7], or the fabrication variation of micro and nanoelectronic circuitry. Since it is difficult for counterfeited ICs to possess the same fingerprints as an authentic one, this technique is successful in counterfeit detection and hardware trojan detection [8]. However, adding unique keys as fingerprints on each IC during manufacturing can be prohibitively expensive. Physically Unclonable Function (PUF)-based fingerprinting provides a better alternative as discussed later.

T4. *Circuit Obfuscation*: Circuit obfuscation and IC camouflaging techniques try to obfuscate the functional properties of a given IC by adding additional logical blocks in the design[9][10][11]. An obfuscated circuit essentially hides the IP and thus protects the key design ideas from being stolen. However, circuit obfuscation techniques usually require a key for proper operation and the security of the key can dictate the success of IP protection [9][10].

T5. *Parametric and Functional Tests*: Parametric and functional tests provide simple but effective techniques for detecting a counterfeit integrated circuit. Parametric tests constitute a comparison of a suspected chip with a golden chip using different measurement techniques such as leakage current measurement, power consumption measurement, threshold current detection, propagation delay testing and set-up time testing. [12][13]. Although parametric tests are time efficient, they suffer from the effects of process variation, IC usage, and aging. Functional tests verify the functionality of the chip, and therefore, do not perform well if the counterfeit product performs all of the functionality as the original. At a given facility these tests are easy to perform and can effectively support all different generations of a given IC.

T6. *Hardware Metering*: Active or passive hardware metering provides a secure way of tracking the usage of a manufactured chip. Passive metering is similar to fingerprinting; however, the use of a specific IC is also monitored [14]. Active metering can access, lock, or unlock a metered IC for controlling its usage [15]. Although hardware metering can be the preferred solution in terms of security, the implementation cost for such a design needs to be carefully considered.

T7. *Physically Unclonable Functions (PUFs)*: PUFs can provide strong challenge-response-based authentication mechanisms for authentic ICs. Since the challenge-response pairs are derived from the unique physical fingerprint of a given device, it is physically impossible to replicate the same responses [16]. As a result, PUFs can provide great security benefits against IC counterfeiting; however, PUFs can lead to significant implementation and deployment costs.

T8. *Aging Models and Sensors*: Finally, aging models and sensor can be used for detecting recycled counterfeit ICs [17].

IV. Conclusion

Although there exist several anti-counterfeiting techniques, the implementations of such techniques are not widely practiced yet. The key reasons for this scenario can be associated with inertia in design and development, cost of maintaining an anti-counterfeit program, lack of technological advancement for testing design and system integrity during runtime, cost of recycling and regulated disposal, cost of maintaining the database of failed and replaced parts, compliance with regulations, and an overall apathy towards counterfeiting.

We provide the following recommendations to business organizations for addressing the IC counterfeiting problem:

a. Manufacturers should implement preferable anti-counterfeiting measures

b. Designers should secure sensitive data and hardware first, and consider fault tolerant electrical designs for critical systems

c. Newer designs should use authentication techniques to reject counterfeited ICs

d. Legacy hardware and components in critical systems and infrastructure must be maintained, regulated, and recycled systematically

Acknowledgment

Md Tanvir Arafin is supported by Philips Internship Program. The authors would like to thank Dr. Kaan Onarlioglu and Dr. Romina Jose for their kind reviews and comments on this work.

References

[1] 2015 situation report on counterfeiting in the European Union. [Online]. Available: https://euipo.europa.eu/ohimportal/documents/11370/80606/2015+Situation+Report+on+Counterfeiting+in+the+EU

[2] Winning the battle against counterfeit semiconductor products. [Online]. Available: http://www.semiconductors.org/clientuploads/Anti-Counterfeiting

[3] M. Tehranipoor and F. Koushanfar, "A survey of hardware trojan taxonomy and detection," *IEEE Design & Test of Computers*, vol. 27, no. 1, 2010.

[4] Supply chain toolkit 2014. [Online]. Available: http://www.ipo.gov.uk/ipctoolkit.pdf

[5] Counterfeit SD card problem is widespread. [Online]. Available: http://www.eetimes.com/document.asp?doc_id=1326059

[6] G. Qu and M. Potkonjak, *Intellectual property protection in VLSI designs: theory and practice*, 2007.

[7] J. A. Roy, F. Koushanfar, and I. L. Markov, "Ending piracy of integrated circuits," *Computer*, vol. 43, no. 10, pp. 30–38, 2010.

[8] D. Agrawal, S. Baktir, D. Karakoyunlu, P. Rohatgi, and B. Sunar, "Trojan detection using IC fingerprinting," in *IEEE Symposium on Security and Privacy,*, 2007, pp. 296–310.

[9] J. Rajendran, Y. Pino, O. Sinanoglu, and R. Karri, "Security analysis of logic obfuscation," in *Annual Design Automation Conference*, 2012, pp. 83–89.

[10] J. Rajendran, M. Sam, O. Sinanoglu, and R. Karri, "Security analysis of integrated circuit camouflaging," in *ACM SIGSAC conference on Computer & communications security*, 2013, pp. 709–720.

[11] J. Zhang, "A practical logic obfuscation technique for hardware security," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 24, no. 3, pp. 1193–1197, 2016.

[12] K. Lofstrom, W. R. Daasch, and D. Taylor, "IC identification circuit using device mismatch," in *Digest of Technical Papers, IEEE International Solid-State Circuits Conference*, 2000, pp. 372–373.

[13] G. F. Nelson and W. F. Boggs, "Parametric tests meet challenge of high-density ICs," *Electronics*, vol. 48, no. 25, pp. 108–111, 1975.

[14] F. Koushanfar, G. Qu, and M. Potkonjak, "Intellectual property metering," in *International Workshop on Information Hiding*. Springer, 2001, pp. 81–95.

[15] Y. Alkabani and F. Koushanfar, "Active hardware metering for intellectual property protection and security," in *USENIX security*. Boston MA, USA, 2007, pp. 291–306.

[16] B. Gassend, D. Clarke, M. Van Dijk, and S. Devadas, "Silicon physical random functions," in *ACM conference on Computer and communications security*, 2002, pp. 148–160.

[17] K. K. Kim, W. Wang, and K. Choi, "On-chip aging sensor circuits for reliable nanometer MOSFET digital circuits," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 57, no. 10, pp. 798–802, 2010.